

5 Vectors and transformation geometry

	Students should be taught to:	Notes
5.1 Vectors	Higher Tier only	
5.2 Transformation geometry	A understand that rotations are specified by a centre and an angle	
	B rotate a shape about a point through a given angle	
	C recognise that an anti-clockwise rotation is a <i>positive</i> angle of rotation and a clockwise rotation is a <i>negative</i> angle of rotation	
	D understand that reflections are specified by a mirror line	Such as $x = 1$, $y = 2$, $y = x$, $y - x = 0$
	E construct a mirror line given an object and reflect a shape given a mirror line	e.g. reflect a triangle in the line $y = x$
	F understand that translations are specified by a distance and direction	
	G translate a shape	
	H understand and use column vectors in translations	
	I understand that rotations, reflections and translations preserve length and angle so that a transformed shape under any of these transformations remains congruent to the original shape	
	J understand that enlargements are specified by a centre and a scale factor	Positive scale factor only (including fractions)
	K understand that enlargements preserve angles and not lengths	
	L enlarge a shape given the scale factor	With or without a centre given
	M identify and give complete descriptions of transformations	